Deep Learning and its Applications to Pollen Analysis

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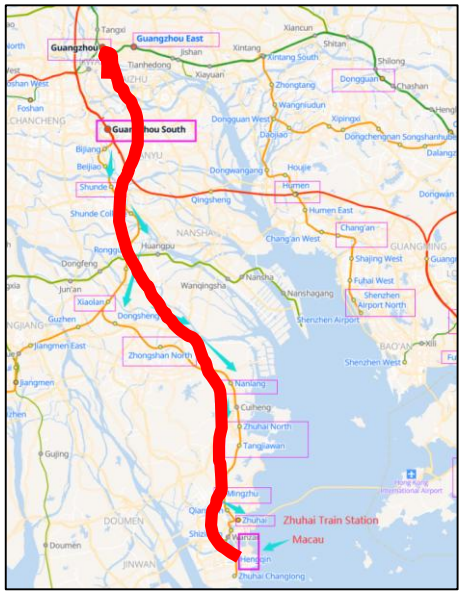
43rd Mid-Continent Paleobotanical Colloquium
Panama 2026

What lead me here?



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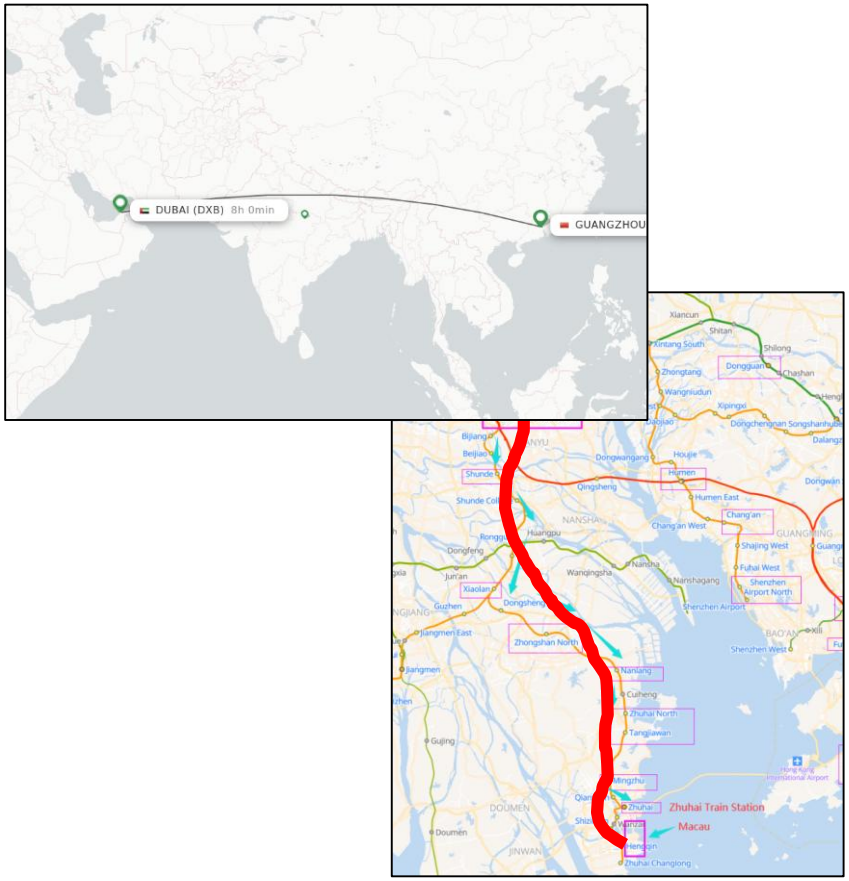

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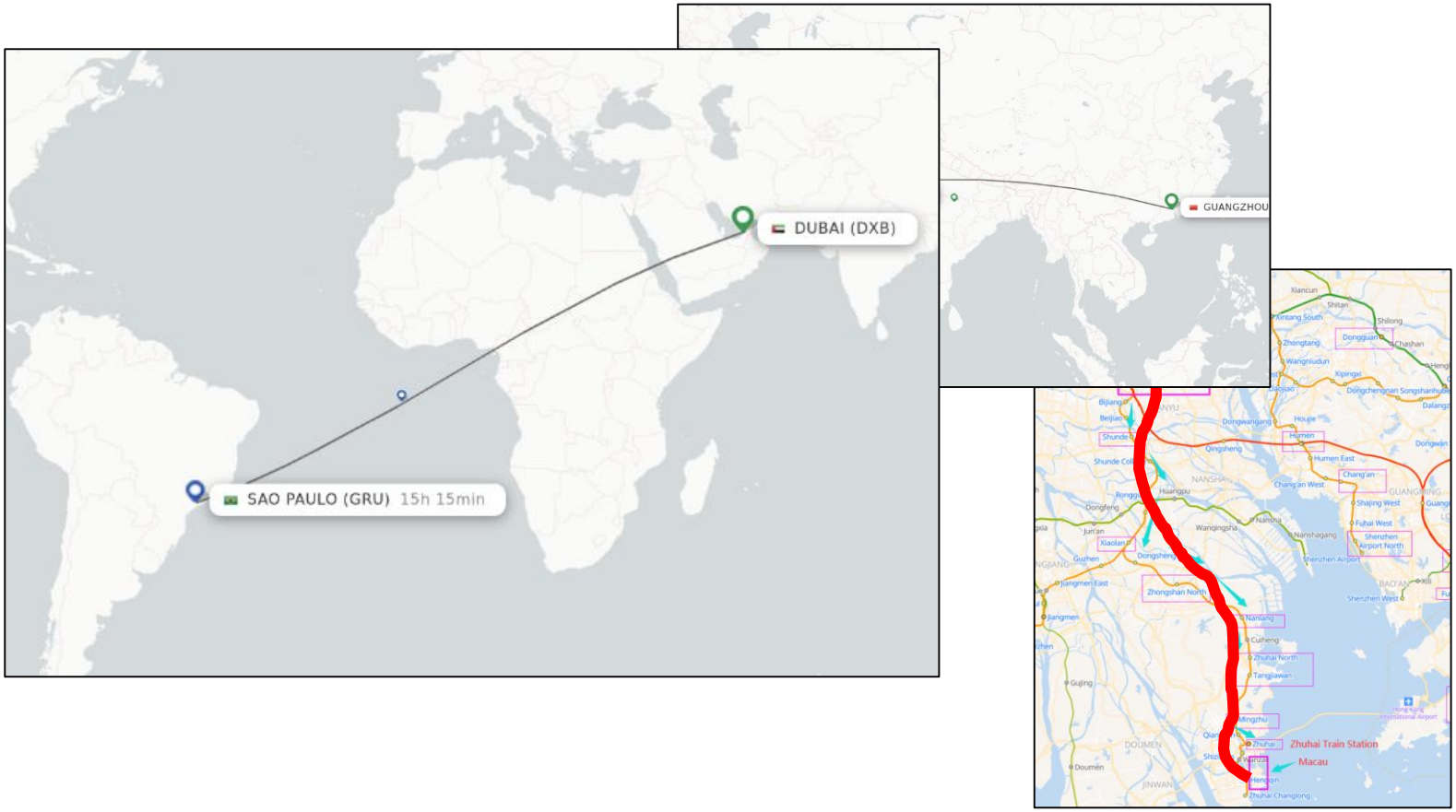

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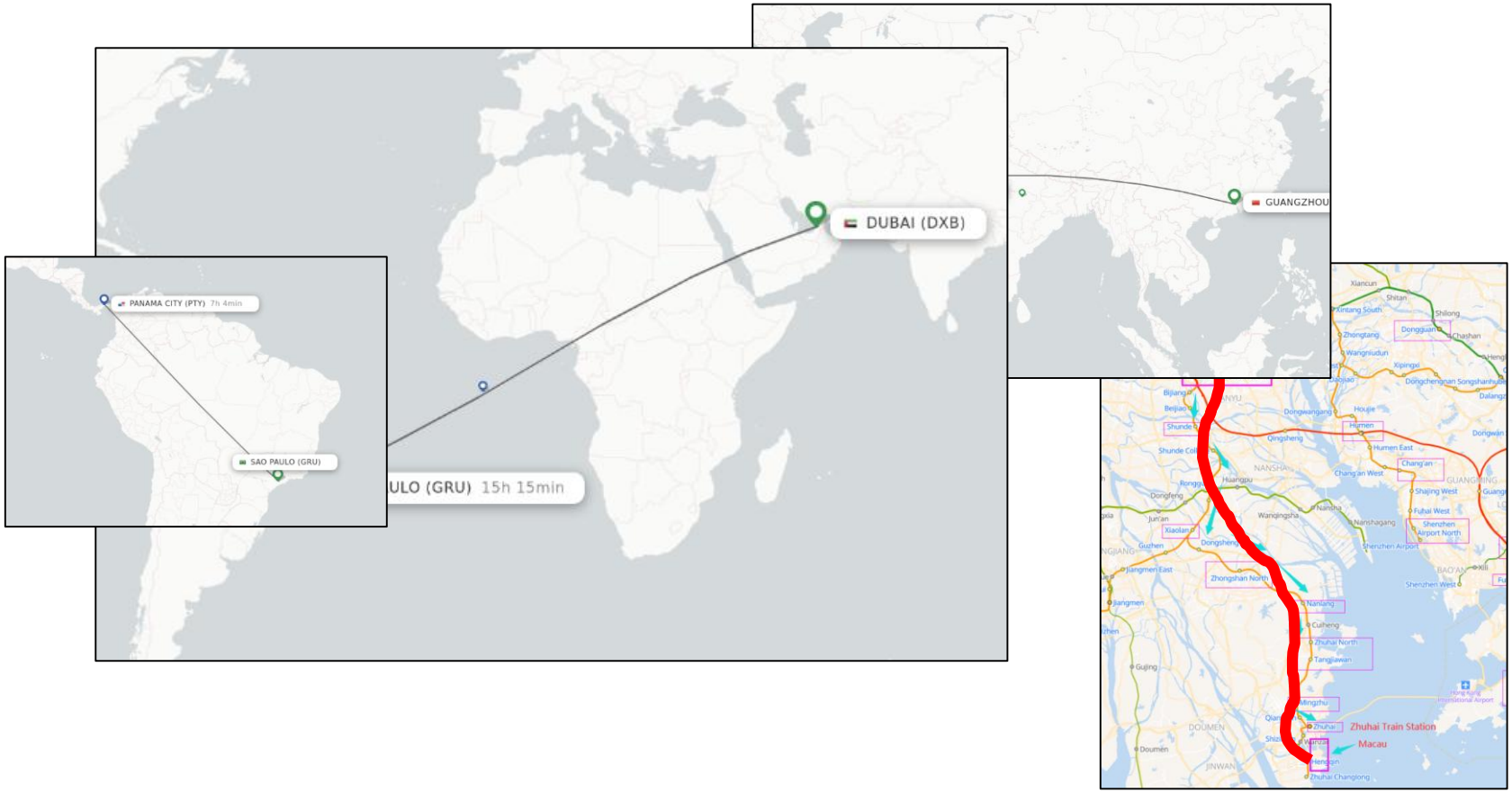
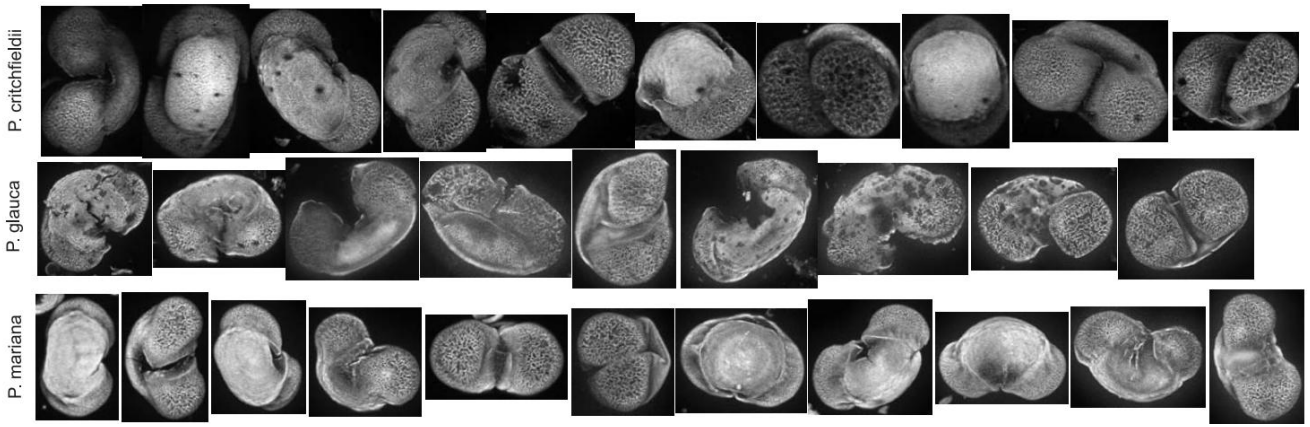
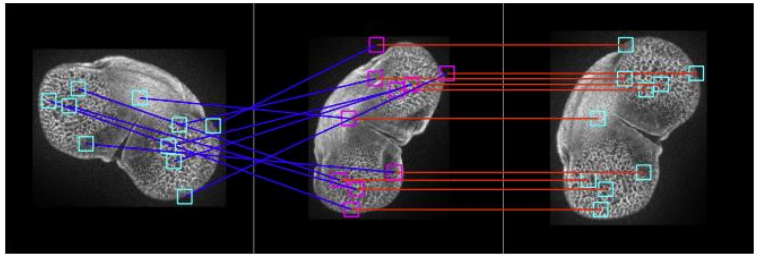


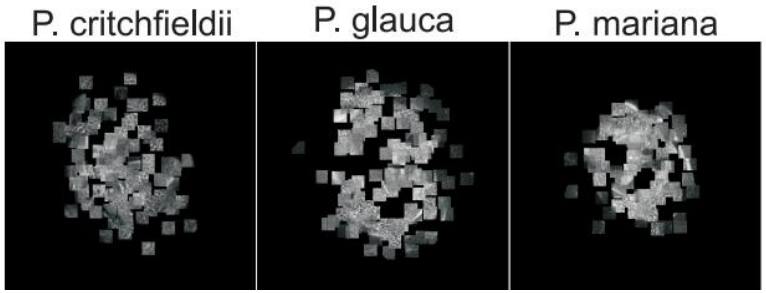
What lead me here?

2014 PhD

Non-deep learning for pollen recognition







Non-deep learning for fossil pollen recognition with Zeiss Apotome fluorescence microscope

S. Kong, S. Punysena, C. Fowlkes, "Spatially Aware Dictionary Learning and Coding for Fossil Pollen Identification", CVPRW, 2016

What lead me here?

2014 PhD — Postdoc

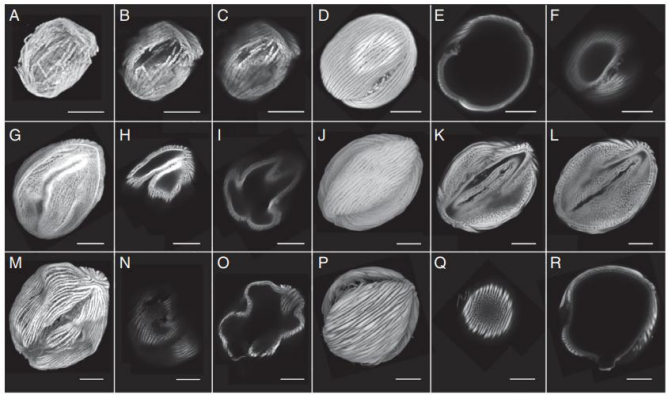
Non-deep learning for pollen recognition

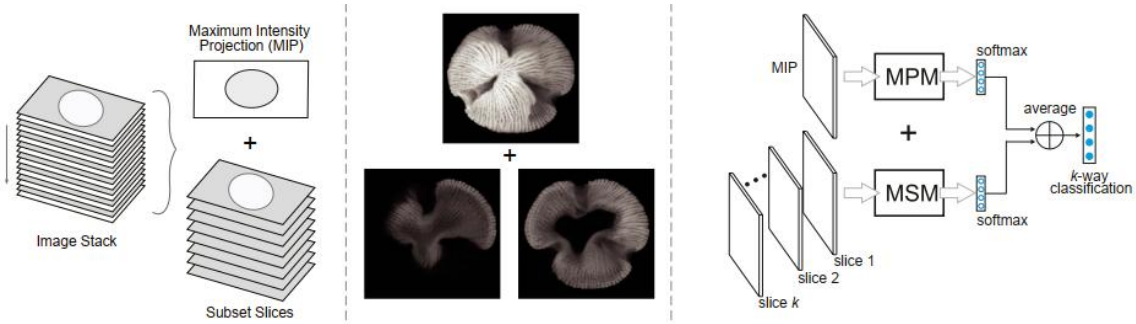
Deep learning for pollen recognition

NSF Stories

Modern computational tools open new era of fossil pollen research

Integrating machine-learning technology with high-resolution imaging helps identify plant species





Deep learning for fossil pollen recognition with Airyscan confocal superresolution microscopy

Romero, et al., "Improving the taxonomy of fossil pollen using convolutional neural networks and superresolution microscopy", PNAS, 2020

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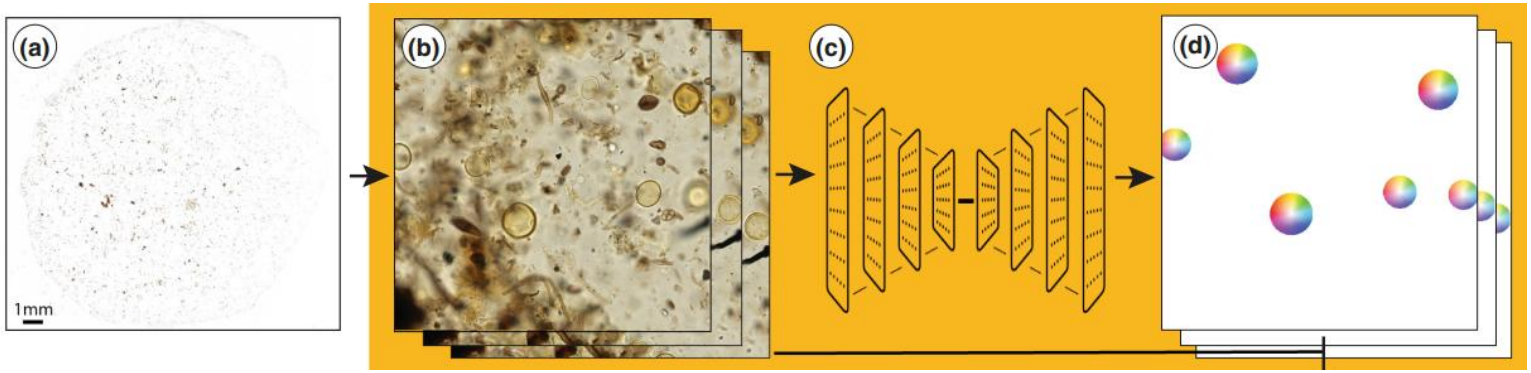
2014 PhD — Postdoc — faculty

Non-deep learning for pollen recognition

Deep learning for pollen recognition

Deep learning for pollen detection



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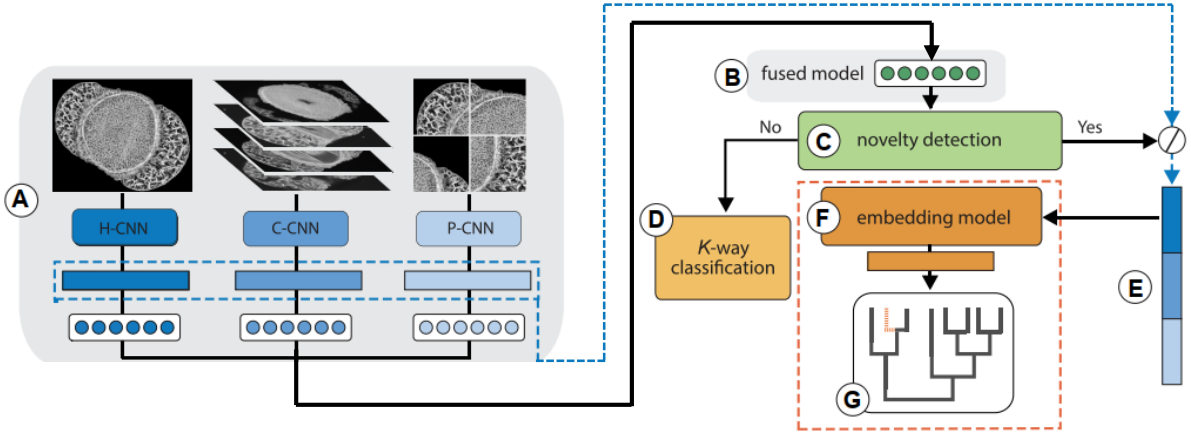
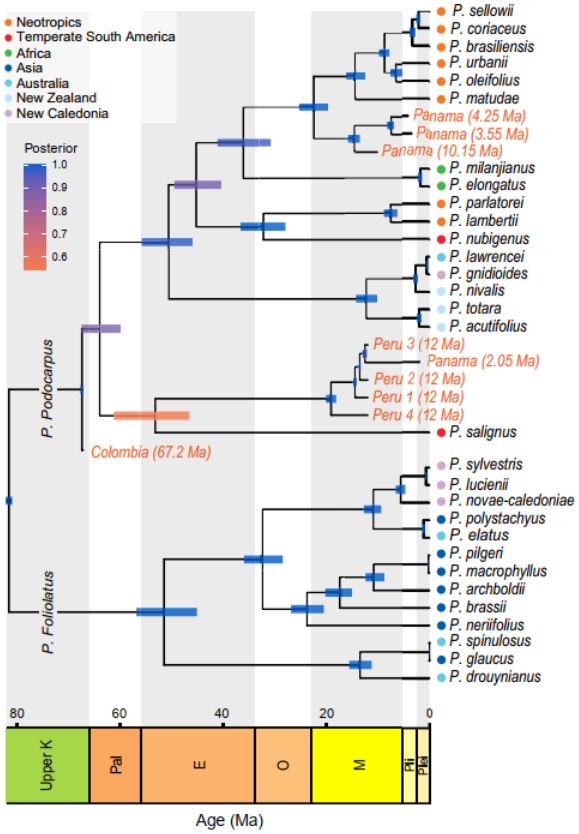
PhD	Postdoc	faculty	faculty
Non-deep learning for pollen recognition	Deep learning for pollen recognition	Deep learning for pollen detection	Deep learning for deep pollen analysis





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What lead me here?

1. Meet people.
2. Do something together.
3. Enjoy a short stay in Panama.

What to get from this short course?

1. *AI –related topics*: computer vision, machine learning, deep learning
2. *Hand-on experience*: image processing, model training, foundation model adaptation
3. A safe venue to discuss interdisciplinary research

Lecture	Topics
1. Introduction	machine learning, computer vision, deep learning, artificial intelligence
2. Basics via Classification	cross entropy, loss function, optimization, feature, learning rate, gradient descent, regularization, etc.
3. Deep Neural Networks	Neural Network, pre-processing, data augmentation, feature learning, pretraining, finetuning
4. Hands-On Practice: Basics	pollen data preparation, visualization, nearest neighbor, linear classifier, pretrained / foundation model
5. Segmentation and Detection	semantic segmentation, instance segmentation, U-Net, proposal detection, foreground segmentation
6. Foundation Models and Finetuning	CLIP, Segment Anything Model, DINOv2, finetuning, model adaptation
7. Advanced Topics	novel species recognition, phylogentic regression, phylogentic reconstruction via clustering
8. Hands-On Practice: Advanced Topics	pollen detection, finetuning foundation models

What we will learn?

- Pollen Recognition
- Pollen Detection
- Novel Taxa Detection
- Phylogenetic Reconstruction

- Image Classification
- Object Detection
- Open-Set Recognition
- Clustering

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Roadmap

Toolboxes:

- platform: Python, Jupyter Notebook
- basics: sklearn, skimage, numpy, scipy, matplotlib
- for deep learning: PyTorch

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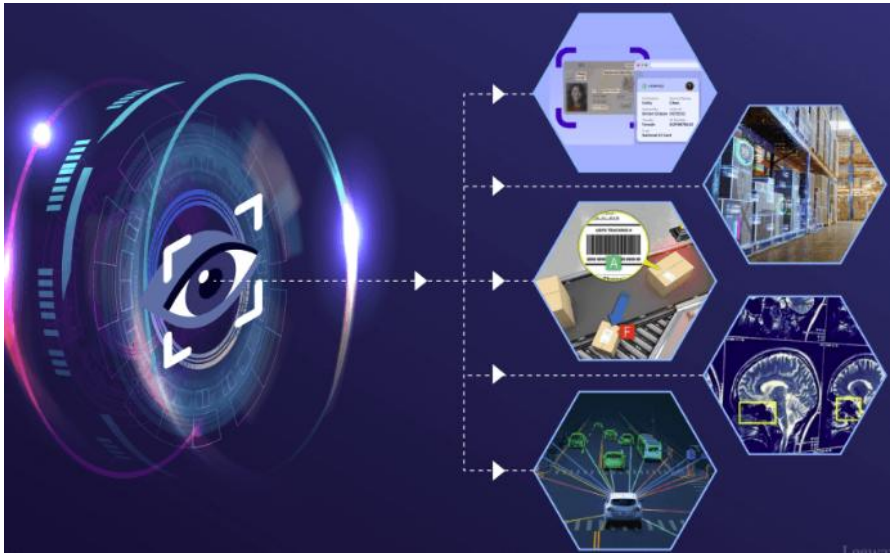
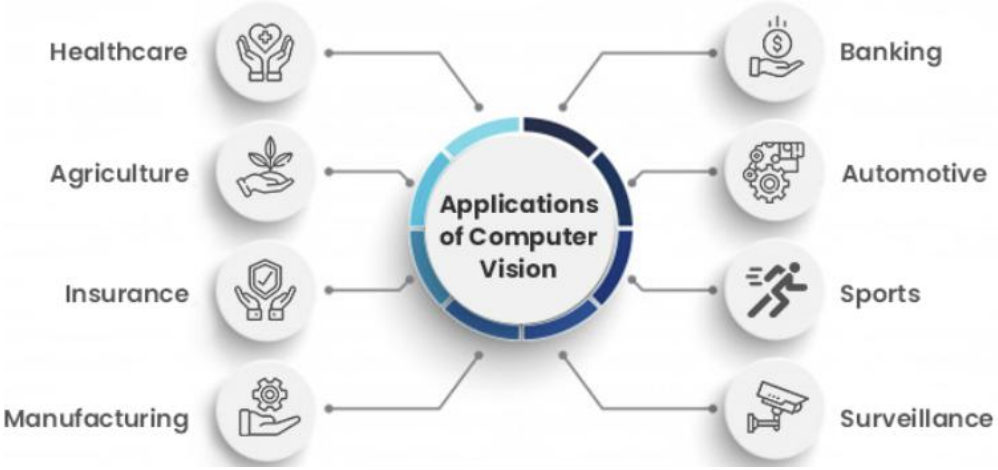
Terminology

Computer Vision (CV) is a field of artificial intelligence (AI) that enables computers to interpret, analyze, and understand *visual data* from the world, such as images and videos, to make decisions or recommendations.





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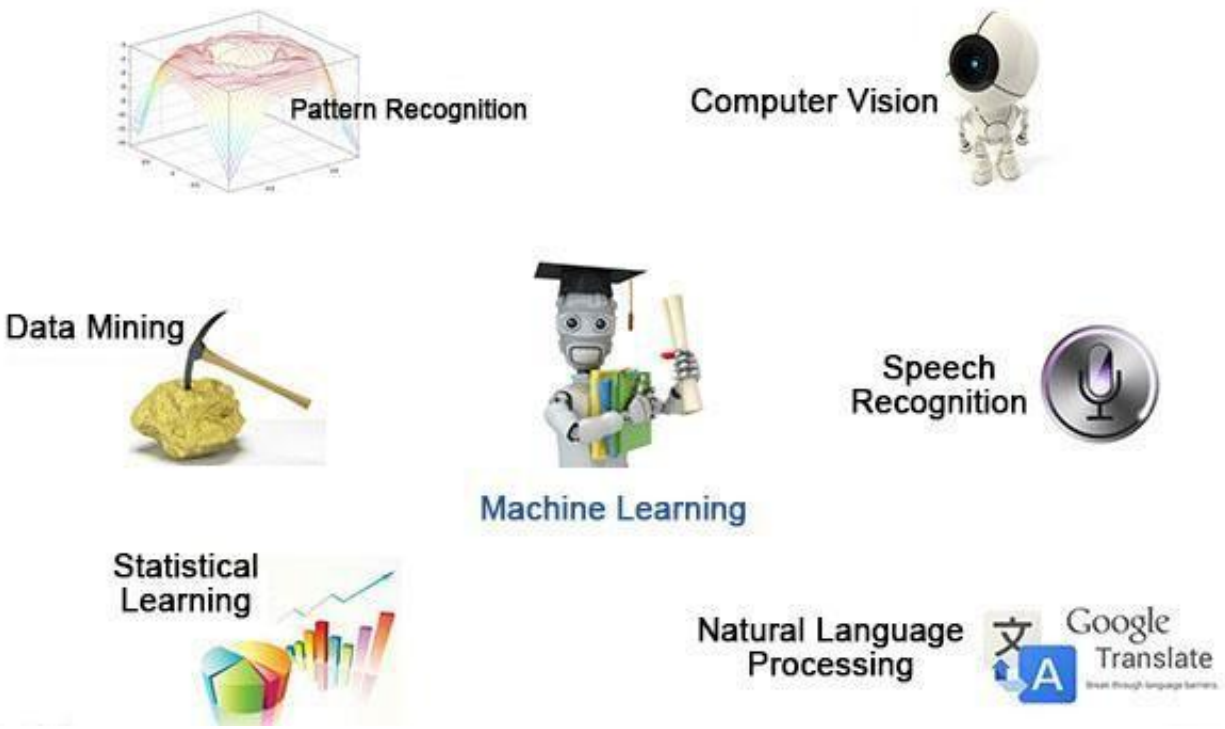

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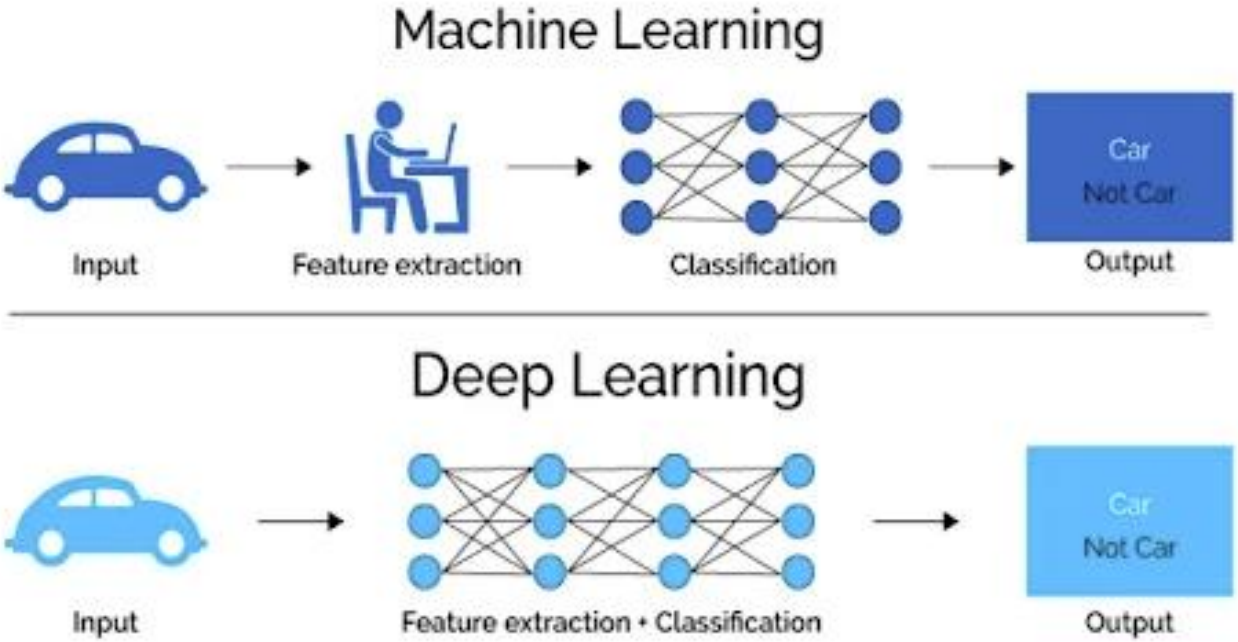

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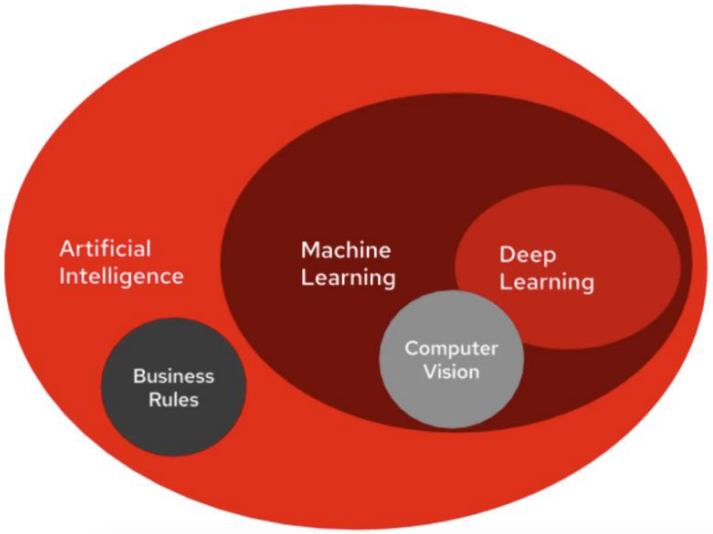
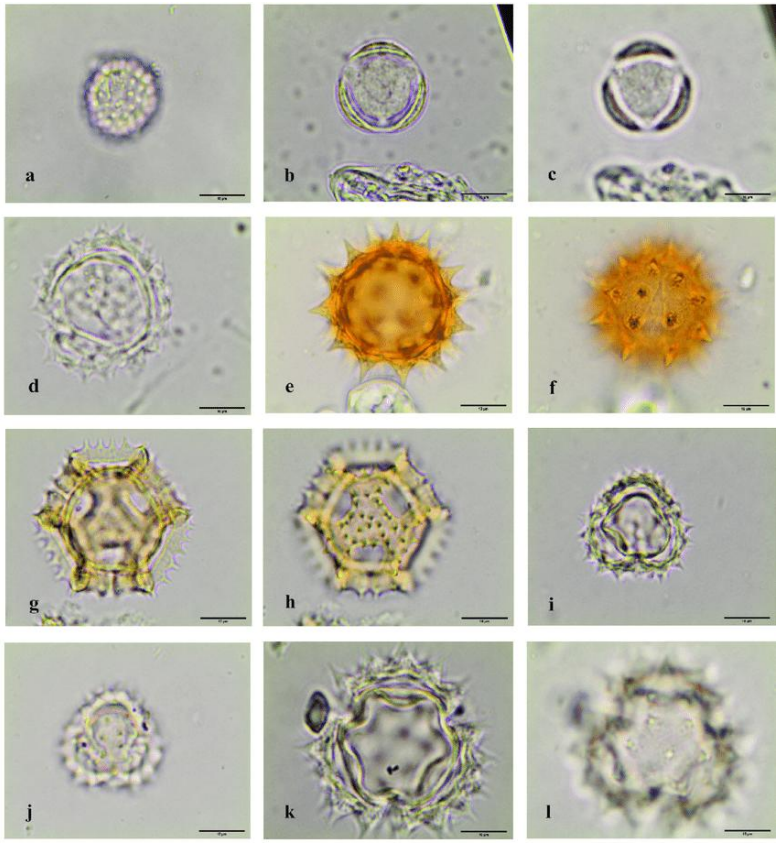
Artificial Intelligence (AI) is the simulation of human intelligence processes by machines, particularly computer systems, enabling them to learn, reason, problem-solve, and perceive.



Pollen Recognition



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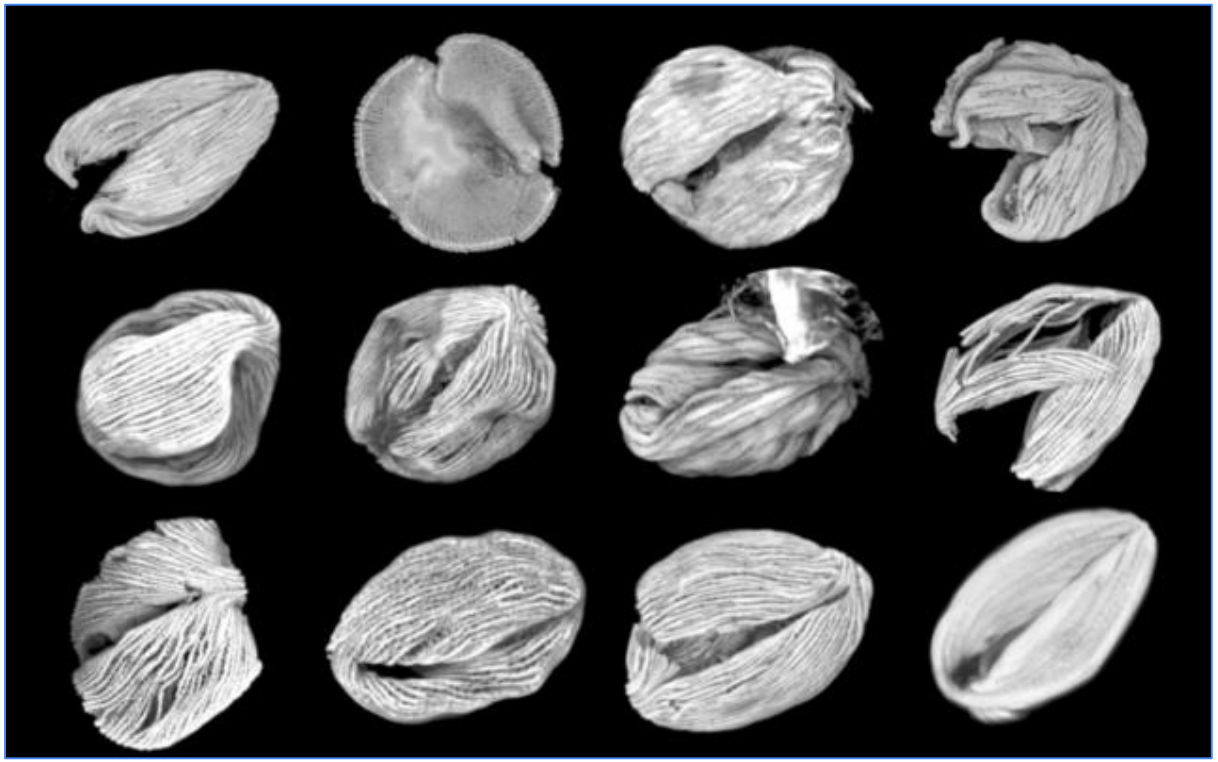

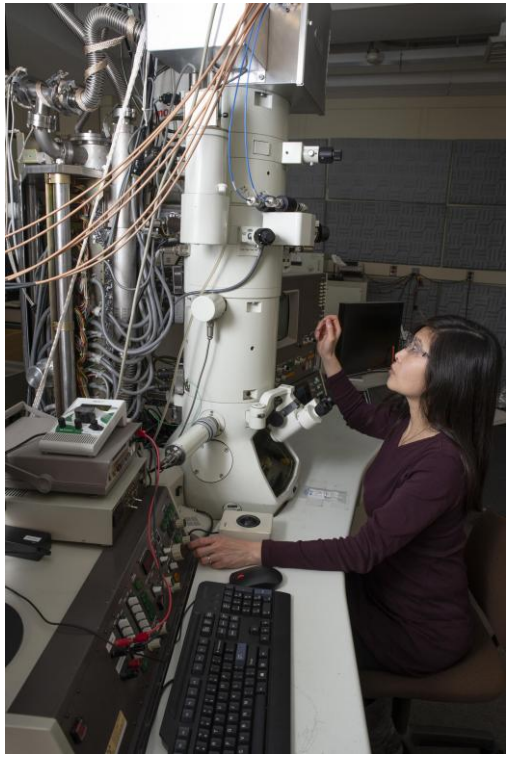

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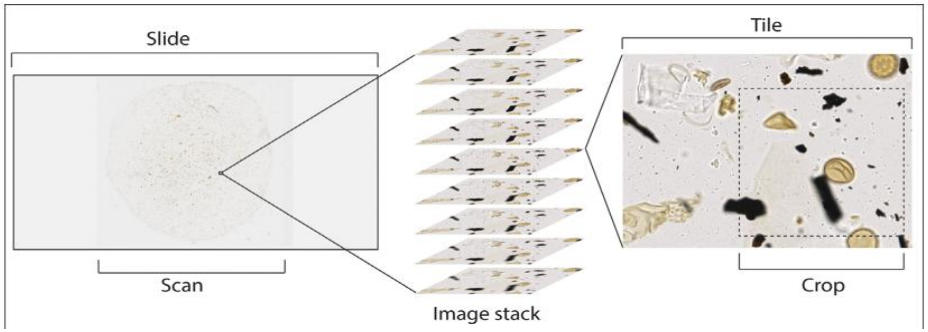
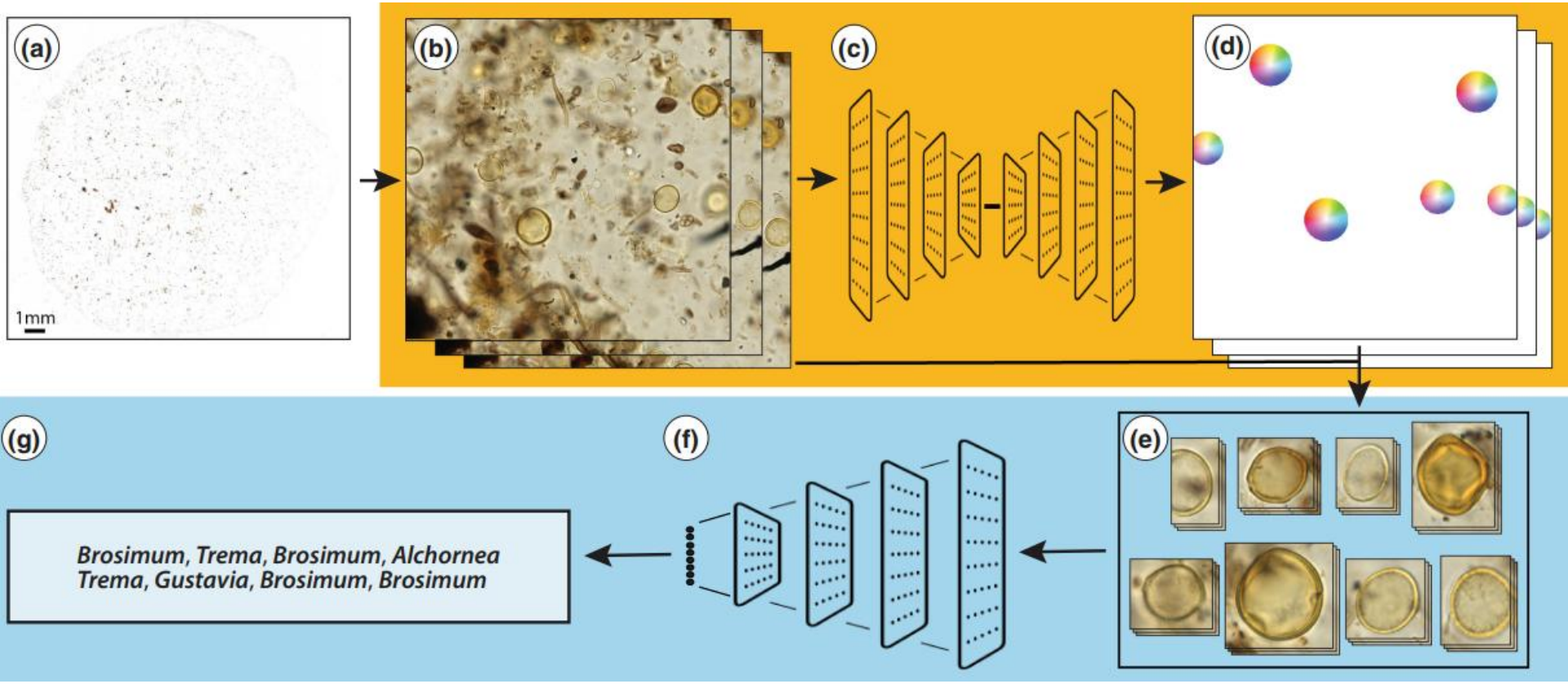
Pollen Detection



A Pipeline: Pollen Detection + Recognition



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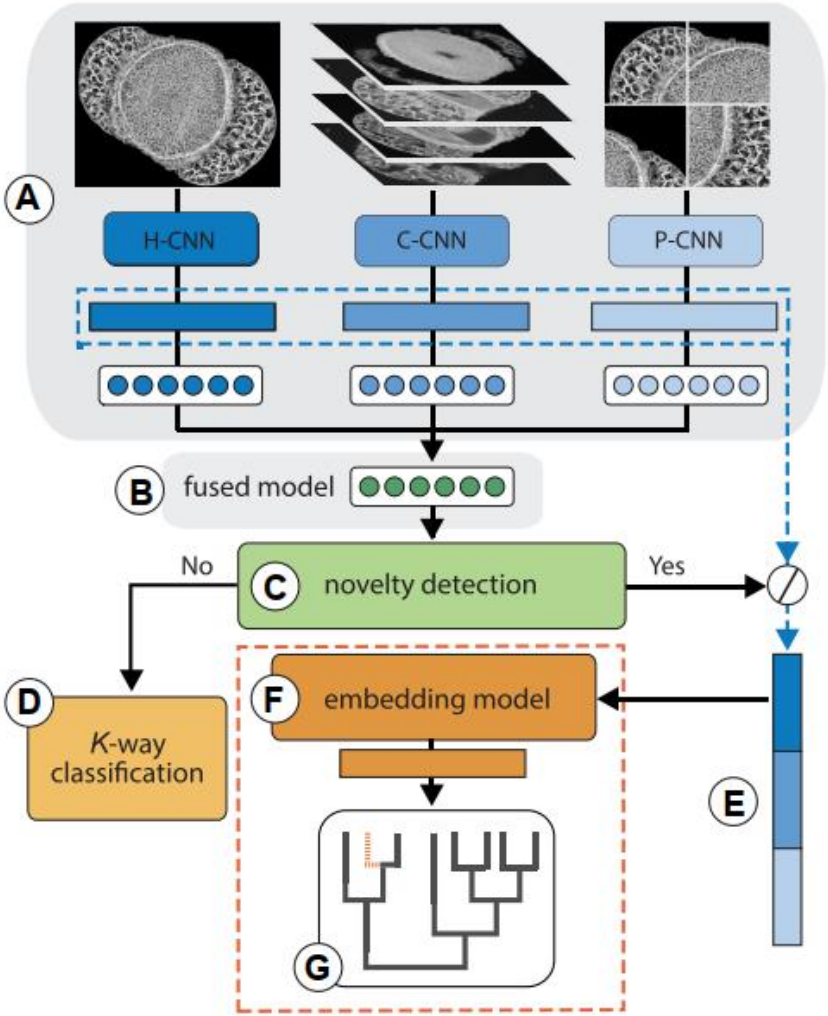
Beyond Recognition: Novel Taxa Detection



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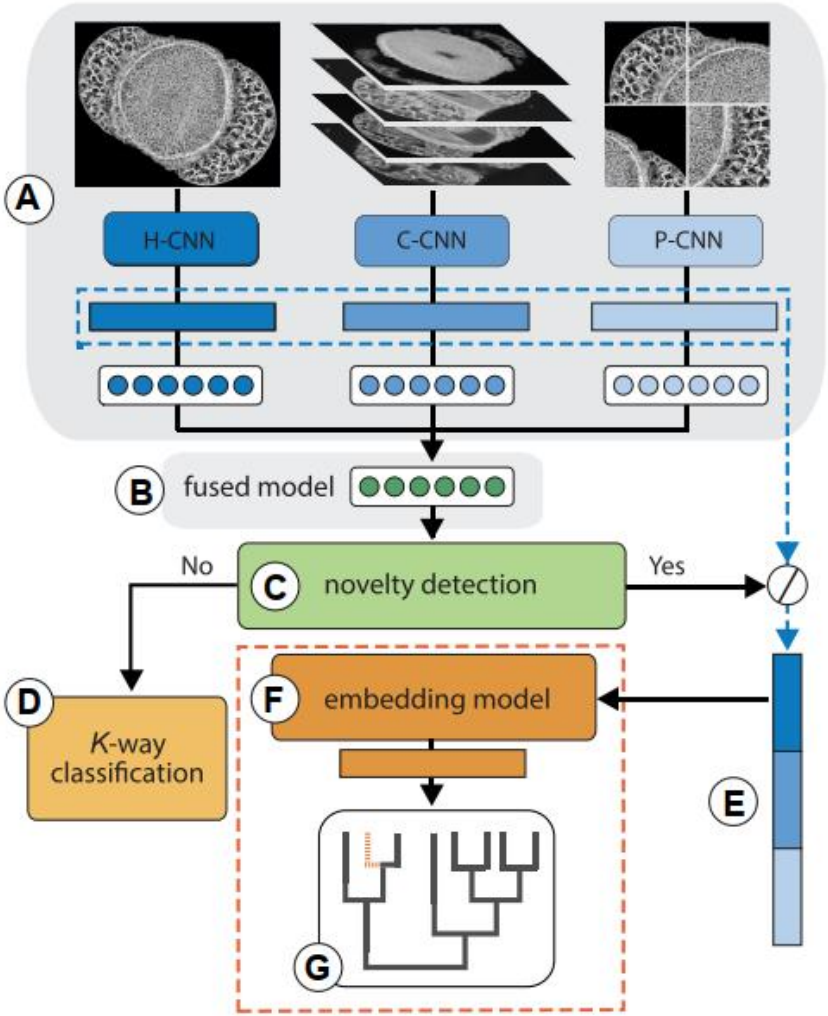
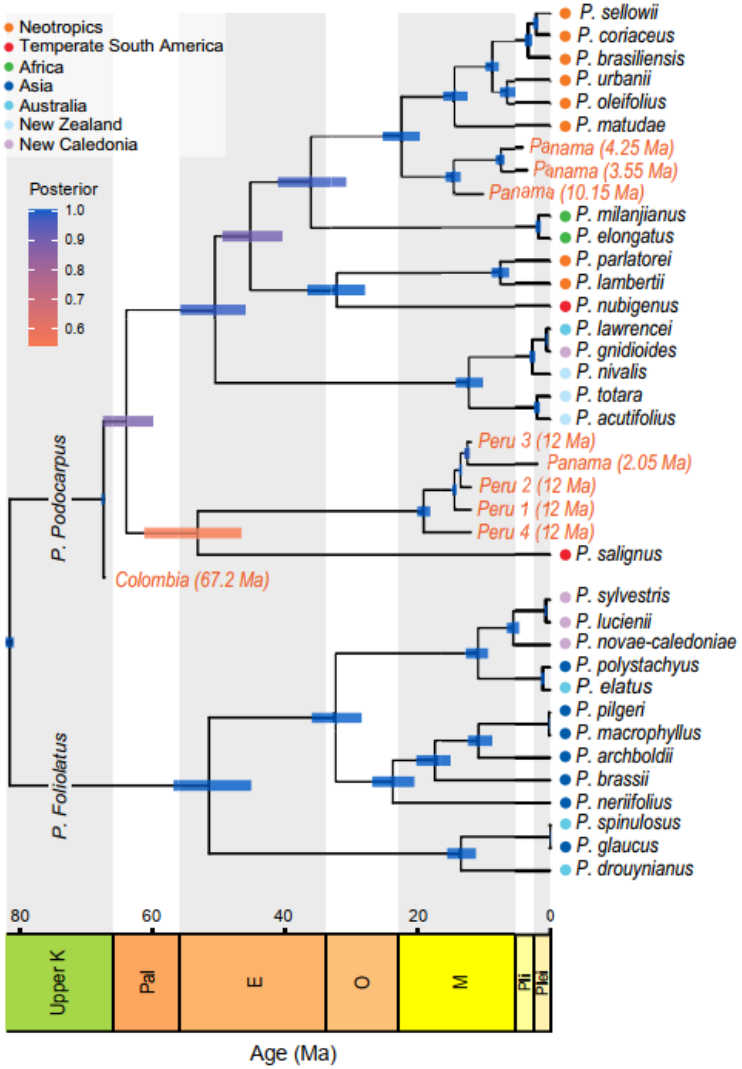
Beyond Recognition: Phylogenetic Reconstruction





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Misc

- Let’s make it interactive.
- Let’s make it comfy.
- Let’s make it continual.